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Research report

Mapping of the progressive metabolic changes occurring during the development of hippocampal sclerosis in a model of mesial temporal lobe epilepsy

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Abstract

We have recently characterized the histopathological changes in an experimental model of mesial temporal lobe epilepsy (MTLE) induced by the intrahippocampal injection of low dose of kainate in mice. Although cerebral metabolism and blood flow are extensively studied and used in human MTLE to locate the regions involved in seizures before surgery, this exploration is only performed once the disease has fully developed. Therefore, in the present study, we followed the temporal evolution of intrahippocampal kainate-induced metabolic changes in mice from kainate injection to 120 days later by the quantitative autoradiographic [^{14}C]2-deoxyglucose (2DG) technique. At day 0 (late phase of status epilepticus (SE)) and 15 days after kainate, i.e., during the period of ongoing neuropathological changes, glucose utilization was decreased bilaterally in all parts of the cerebral cortex, and ipsilaterally in the thalamus. In the hippocampus, CA1 metabolic activity was depressed at day 0 and increased at day 15 while CA3 glucose utilization was increased at both day 0 and 15. By day 30, there were almost no pyramidal cells left in the two hippocampal regions. At day 120, ipsilateral decreases persisted in the entorhinal cortex, anterior and ventromedian thalamus, and metabolic increases were recorded bilaterally in the central amygdala, anterior hypothalamus and mamillary body. At all times after kainate, a normo-, hypo- or hypermetabolic level was recorded in the dentate gyrus. The present study shows that the process of hippocampal sclerosis involves bilateral cortical reactivity and the participation of some limbic forebrain and motor structures. When hippocampal sclerosis has fully developed, hypometabolism is limited to regions directly connected to the damaged hippocampus and most likely involved in the new hyperexcitable circuit of limbic seizures.
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1. Introduction

Human mesial temporal lobe epilepsy (MTLE), by far the most common clinical form of epilepsy [14,24,25] usually appears in adolescent or adult subjects that have undergone complicated febrile convulsions, status epilepticus (SE), encephalitis or head trauma in early childhood [12]. After several asymptomatic years, seizures occur spontaneously [13]. The MTLE syndrome is characterized by hippocampal sclerosis consisting of an extensive loss of pyramidal cell bodies of the CA1 and CA3 areas, selective loss of NPY- and somatostatin-containing cells in the hilus of the dentate gyrus, reactive gliosis and mossy fiber reorganization, associated in about 50% of the cases with granule cell dispersion in the dentate gyrus. These pathological patterns are quite consistently found in surgically-treated human hippocampal epilepsy [9,13,21,23,26]. Patients with marked hippocampal sclerosis have a characteristic clinical history and post-surgical prognosis, well-localized electroencephalographic (EEG) abnormalities, neuropsychological deficits and altered metabolic patterns [13].

In MTLE, seizures often become medically intractable and a proportion of patients undergoes the surgical resec-

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tion of epileptic foci after a detailed clinical examination including a positron emission tomography (PET) or a single photon emission computed tomography (SPECT) scan allowing to locate the extent of the functional disturbances involving the epileptic focus [13,30,43]. The [^{18}F]fluorodeoxyglucose (FDG)-PET scan demonstrates widespread interictal hypometabolism which commonly involves a much larger area than the pathological substrate, i.e., hippocampal sclerosis, including the entire temporal lobe and occasionally the ipsilateral thalamus, basal ganglia and some neocortical areas [20,35–38,41]. A characteristic ictal and postictal SPECT pattern has also been described for MTLE: unilateral temporal hyperperfusion during the ictal period, lateral temporal hypoperfusion but mesial hyperperfusion in the immediate postictal period and complete temporal hypoperfusion in the interictal phase [32].

We have recently characterized and validated an experimental model of MTLE secondary to the intrahippocampal injection of a small dose of kainate in mice. This model reproduces most of the features of human MTLE. Specifically, we record acute cell loss in the dentate gyrus hilus. Thereafter, major neuronal loss occurs progressively and consistently in the CA1 hippocampal area paralleled by granule cell dispersion, astrogliosis and mossy fiber sprouting. As in human MTLE, recurrent partial seizures do not change over time, remaining limited to limbic structures and being very discrete. The only difference between human MTLE and the mouse model is the fact that recurrent seizures occur in the mice immediately after the SE phase without any silent period [5,6].

Despite the extensive studies performed in humans, the metabolic or circulatory pattern of hippocampal sclerosis and MTLE is only analyzed when the disease has already fully developed, thus not allowing any insight in the mapping of the progressive functional changes underlying the development of MTLE. Indeed, such a temporal follow-up would allow to better understand the mechanisms underlying the constitution of hippocampal sclerosis and to eventually develop prevention strategies. Therefore, in the present study, by means of the quantitative autoradiographic [^{14}C]2-deoxyglucose (2DG) method, we explored the metabolic changes occurring from the induction of SE induced by the intrahippocampal injection of kainate to mice until 4 months later in order (i) to map the progressive involvement of the various brains areas during the development of hippocampal sclerosis and (ii) to characterize further this experimental model of MTLE.

2. Materials and methods

2.1. Animals and surgical procedure

Experiments were performed on 23 adult male Swiss albino mice (CERJ, Le Genest-St-Isle, France) weighing 30–35 g. They were housed under standard laboratory conditions on a 12/12 h light/dark cycle (lights on at 0700 h). All animal experimentation was performed in accordance with the rules of the European Committee Council Directive of November 24, 1986 (86/609/EEC) and the French Department of Agriculture (Licence No. 00733).

Mice were injected either with saline or with kainate into the dorsal hippocampus (coordinates from bregma: $A = -1.8$ mm and $L = -1.8$ mm, from brain surface: $H = -1.8$ mm) under equitesin anesthesia (4 ml/kg), as previously described [5,42]. Kainate, dissolved in saline (50 nl, 0.2 μg), was injected over 5–10 min through a metal injection cannula of 100 μm diameter using a Hamilton syringe. After the injection, the capillary was maintained in situ for an additional 2 min to limit reflux along the injection track. The kainate injection induced the rapid occurrence of SE in mice at awakening from anesthesia. SE lasted for a mean duration of 5–10 h and was characterized by continuous spikes and spike-and-waves on the EEG.

2.2. Measurement of local cerebral metabolic rates for glucose (LCMRglcs)

A femoral artery and vein were catheterized with polyethylene under light halothane anesthesia. Both catheters were threaded under the skin, up to the back of the neck to allow free access to the catheters without disturbing mouse movements. The animals were allowed to recover from surgery in their home cages, for at least 3–4 h before the onset of the experiment. The autoradiographic experiments were performed on a total number of 23 mice, eight receiving saline (two at each time) and 15 receiving kainate.

LCMRglcs were measured by the [^{14}C]2DG technique [39] adapted to small rodents [28]. The measurement of LCMRglcs was performed at 10–12 h after kainate injection, i.e., at the end of SE, considered as day 0, and at 15, 30 and 120 days after the injection of kainate or saline to the mice. The [^{14}C]2DG (4.625 MBq/kg, specific activity 1.65–2.04 GBq/mmol, Isotopchim, Ganagobie-Perhuis, France) was injected as an i.v. pulse to freely moving mice. Timed arterial blood samples were drawn over the following 45 min for the measurement of plasma glucose and 2DG concentrations. At approximately 45 min after the injection of the tracer, the animals were sacrificed by decapitation. Brains were rapidly removed, frozen and cut into 20-μm coronal sections. Sections were autoradiographed on Amersham Biomax MR film along with [^{14}C]methylmethacrylate standards calibrated for their ^{14}C concentration in brain tissue. The autoradiographs were then digitized and analyzed by densitometry with an image processing system (Biocom 500, Les Ulis, France). The location of specific nuclei was assessed on adjacent sections stained with cresyl violet. LCMRglcs were calculated

according to the operational equation of the 2DG method [39].

2.3. Measurement of physiological variables

Mean arterial blood pressure was regularly recorded over the experimental procedure with an air-damped mercury manometer. Just prior to the 2DG injection and sacrifice, samples of arterial blood were taken for the measurement of blood pH, pO_2 and pCO_2 by means of a blood gas analyzer (Corning, model 158, Corning Medical and Scientific, Halstead, UK). The hematocrit was determined from two arterial blood samples taken just before and 30 min after the 2DG injection.

2.4. Statistical analysis

LCMRglcs were determined in 24 structures both ipsi- and contralaterally to the site of injection in two groups of 3–4 animals at each time after the injection, saline- and kainate-treated mice. Values of LCMRglcs in the saline group were homogeneous whatever the time of measurement after the injection. Therefore, to obtain group number homogeneity, one saline-treated mouse was randomly selected at each of the four times studied to constitute the saline-exposed group. The values of LCMRglcs and physiological variables recorded in the kainate-exposed groups were compared to those in the saline group by means of two-way analysis of variance (ANOVA) followed by a Dunnett's *t*-test for multiple comparisons.

3. Results

3.1. Behavior

The behavior of the saline-exposed mice was not altered by the injection. The behavior of kainate-treated mice was similar to the one described previously [5]. Briefly, at awakening from kainate injection, mice were immobile most of the time and presented intermittent contraversive clonic deviations of the head and chewing. These symptoms lasted for most of the SE. During the following days, the behavior of the mice was only slightly altered and characterized by episodes of immobility, staring and behavioral arrest, sometimes associated with facial movements. Rare motor seizures characterized by behavioral arrest, vocalization, chewing, clonic head movements and rearing occurred at a frequency of 1–2 per week. These behavioral patterns did not change over the 120 days of the study, except for the progressive appearance of anxiety and aggressivity patterns when the mice were handled.

3.2. Physiological variables

The kainate injection did not affect most of the physiological variables recorded in the present study (Table 1). Only plasma glucose concentration was significantly decreased at the end of the 2DG experiment in the group studied at day 0 and significantly increased during the whole experimental procedure at day 15.

3.3. Local cerebral metabolic rates for glucose

At day 0, i.e., at about 12 h after the acute injection of kainate corresponding to the late phase of SE, LCMRglcs were decreased both ipsi- and contralaterally throughout the cerebral cortex (Table 2). LCMRglc was decreased ipsilaterally to the site of injection in the hippocampal CA1 area and increased ipsilaterally in the lateral septum and hippocampal CA3 layer.

At day 15 after kainate, bilateral metabolic decreases were still recorded in all regions of the cerebral cortex while an ipsilateral decrease in LCMRglc occurred in the ventromedian thalamus. Conversely, LCMRglc was increased in the ipsilateral hippocampal CA1 area of all mice.

Table 1
Physiological variables in control and kainate-injected mice

	Control ($n = 4$)	KA day 0 ($n = 3$)	KA day 15 ($n = 4$)	KA day 30 ($n = 4$)	KA day 120 ($n = 4$)
Plasma glucose level (g/l)					
Zero time	1.86 ± 0.22	1.74 ± 0.39	2.15 ± 0.14*	1.66 ± 0.10	1.68 ± 0.25
45 min after 2DG	1.75 ± 0.14	1.33 ± 0.24**	2.06 ± 0.36*	1.69 ± 0.22	1.69 ± 0.38
Arterial blood pressure (mmHg)					
Zero time	106 ± 5	101 ± 2	102 ± 7	105 ± 7	100 ± 3
20 min after 2DG	102 ± 9	97 ± 2	96 ± 5	100 ± 10	100 ± 8
Arterial pO_2	93 ± 6	88 ± 4	92 ± 9	94 ± 7	94 ± 3
Arterial pCO_2	31 ± 4	31 ± 5	33 ± 4	34 ± 3	32 ± 6
Arterial pH	7.45 ± 0.05	7.42 ± 0.05	7.40 ± 0.04	7.46 ± 0.03	7.43 ± 0.06
Hematocrit (%)	51 ± 4	52 ± 4	51 ± 3	50 ± 3	48 ± 4

* $p < 0.05$, ** $p < 0.01$, statistically significant differences from control.

Table 2
Effects of intrahippocampal kainate on LCMRglcs of adult mice

	Control ($n = 4$)	KA			
		Day 0 ($n = 3$)	Day 15 ($n = 4$)	Day 30 ($n = 4$)	Day 120 ($n = 4$)
Cerebral cortex					
Prefrontal, right	122 ± 12	98 ± 10*	100 ± 9*	103 ± 8	114 ± 7
Prefrontal, left	121 ± 12	97 ± 9*	100 ± 7*	104 ± 8	117 ± 9
Frontal, right	108 ± 9	91 ± 11*	89 ± 10*	92 ± 8	95 ± 14
Frontal, left	105 ± 11	86 ± 11*	91 ± 7*	91 ± 9	93 ± 15
Piriform, right	116 ± 12	89 ± 10*	99 ± 13	95 ± 13	99 ± 11
Piriform, left	112 ± 13	83 ± 11*	93 ± 11*	93 ± 9	100 ± 9
Sensorimotor, right	119 ± 8	98 ± 10*	88 ± 7*	98 ± 8	105 ± 14
Sensorimotor, left	122 ± 10	95 ± 14*	93 ± 6*	102 ± 5	114 ± 9
Entorhinal, right	76 ± 9	60 ± 6*	53 ± 9**	54 ± 13*	56 ± 10*
Entorhinal, left	67 ± 3	61 ± 7	55 ± 4	58 ± 16	57 ± 10
Forebrain					
Medial septum, right	69 ± 9	66 ± 8	54 ± 17	65 ± 10	76 ± 6
Medial septum, left	70 ± 13	65 ± 8	56 ± 14	70 ± 8	79 ± 8
Lateral septum, right	66 ± 13	98 ± 20*	45 ± 14	59 ± 16	63 ± 9
Lateral septum, left	63 ± 14	64 ± 8	46 ± 12	60 ± 11	63 ± 5
Medial amygdala, right	52 ± 4	52 ± 5	44 ± 9	40 ± 7	45 ± 5
Medial amygdala, left	48 ± 8	50 ± 7	43 ± 10	39 ± 6	46 ± 9
Central amygdala, right	44 ± 12	46 ± 9	39 ± 10	36 ± 6	60 ± 7*
Central amygdala, left	40 ± 10	41 ± 12	37 ± 11	36 ± 5	68 ± 3**
Basolateral amygdala, right	82 ± 6	75 ± 3	69 ± 12	62 ± 4	71 ± 7
Basolateral amygdala, left	79 ± 16	62 ± 11	71 ± 8	70 ± 9	80 ± 1
Hippocampal CA1 area, right	80 ± 12	46 ± 4**	120 ± 23**	destroyed	destroyed
Hippocampal CA1 area, left	82 ± 14	71 ± 13	107 ± 19	66 ± 17	65 ± 14
Hippocampal CA3 area, right	91 ± 13	191 ± 47**	112 ± 23	destroyed	destroyed
Hippocampal CA3 area, left	84 ± 13	99 ± 10	106 ± 27	89 ± 11	79 ± 16
Caudate nucleus, right	127 ± 12	121 ± 14	112 ± 10	100 ± 14	126 ± 25
Caudate nucleus, left	122 ± 19	107 ± 16	110 ± 10	103 ± 7	125 ± 18
Hypothalamus					
Anterior, right	54 ± 6	64 ± 5	53 ± 7	54 ± 5	68 ± 11*
Anterior, left	49 ± 8	59 ± 4	52 ± 7	52 ± 6	66 ± 10*
Ventromedian, right	50 ± 11	56 ± 9	43 ± 12	37 ± 4	44 ± 1
Ventromedian, left	46 ± 11	55 ± 10	44 ± 7	37 ± 3	45 ± 6
Posterior, right	79 ± 25	73 ± 11	74 ± 10	81 ± 22	79 ± 19
Posterior, left	77 ± 25	70 ± 9	71 ± 12	77 ± 21	76 ± 18
Mammillary body, right	155 ± 12	131 ± 8	144 ± 10	144 ± 18	194 ± 14**
Mammillary body, left	157 ± 13	130 ± 7	145 ± 8	146 ± 19	187 ± 20*
Thalamus					
Anterior, right	130 ± 5	138 ± 4	116 ± 15	119 ± 17	94 ± 16*
Anterior, left	129 ± 12	106 ± 10	120 ± 15	123 ± 8	104 ± 19
Ventromedian, right	120 ± 12	109 ± 14	91 ± 18*	88 ± 16*	92 ± 12*
Ventromedian, left	116 ± 18	94 ± 17	95 ± 19	107 ± 20	117 ± 12
Posterior, right	109 ± 7	108 ± 14	91 ± 19	89 ± 23	105 ± 26
Posterior, left	111 ± 6	90 ± 11	95 ± 17	106 ± 16	125 ± 22
Midbrain–brainstem					
Substantia nigra pars reticulata, right	70 ± 11	65 ± 7	59 ± 8	57 ± 14	72 ± 10
Substantia nigra pars reticulata, left	66 ± 6	59 ± 5	55 ± 10	56 ± 15	67 ± 9
Substantia nigra pars compacta, right	109 ± 10	85 ± 5*	85 ± 8*	95 ± 17	96 ± 6
Substantia nigra pars compacta, left	105 ± 5	84 ± 2*	84 ± 9**	94 ± 14	103 ± 7
Superior colliculus, right	92 ± 9	68 ± 7	94 ± 11	90 ± 11	108 ± 20
Superior colliculus, left	93 ± 6	69 ± 10	97 ± 15	92 ± 7	106 ± 16

Table 2 (continued)

	Control ($n = 4$)	KA			
		Day 0 ($n = 3$)	Day 15 ($n = 4$)	Day 30 ($n = 4$)	Day 120 ($n = 4$)
White matter					
Genu of the corpus callosum, right	41 ± 12	37 ± 15	26 ± 8	31 ± 5	31 ± 9
Genu of the corpus callosum, left	38 ± 9	35 ± 14	25 ± 7	29 ± 6	34 ± 10

$^{*}p < 0.05$, $^{**}p < 0.01$, statistically significant differences from control group.
Rates of cerebral glucose utilization, expressed as μmol/100 g/min, represent means ± S.D.

At day 30, metabolic levels were decreased ipsilaterally in the entorhinal cortex, and the ventromedian thalamus. From day 30, due to the severe cell damage occurring in the hippocampal CA1 and CA3 areas, no reading was performed in these zones.

At day 120, ipsilateral decreases persisted in the entorhinal cortex, anterior and ventromedian thalamus. Metabolic increases were recorded bilaterally in the central amygdala, anterior hypothalamus and mamillary body.

Finally, at all times after kainate injection, a normo- (four mice), hypo- (four mice) or hypermetabolic level (seven mice) was recorded in the dentate gyrus (Fig. 1). The lowest level of glucose utilization recorded was 27 μmol/100 g/min, the highest one 179 μmol/100 g/min compared to 58 μmol/100 g/min in control mice.

The results of the ANOVA were significant for time and KA treatment (p values ranging from 0.04 to 0.0001), ipsilaterally in the entorhinal cortex, hippocampal CA1 and CA3 regions, anterior hypothalamus, lateral septum and substantia nigra pars compacta, and bilaterally in the superior colliculus and mamillary body.

4. Discussion

In our previous study characterizing this new model of temporal lobe epilepsy induced by the intrahippocampal injection of kainate in mice, we were able to distinguish two phases during the evolution of the histopathological damage. The first one starts from about 2 h after the injection when we could already see some silver-stained damaged neurons and lasts up to 15 days. During this early phase, neuronal damage remains limited. After the first 2 weeks post kainate, the radial dispersion of granule cells starts to reach its maximal extension by 30 days and simultaneously, the process of degeneration of CA1 pyramidal cells is accelerating, spreading to CA3 and becoming nearly complete by 30 days. We hypothesized that this

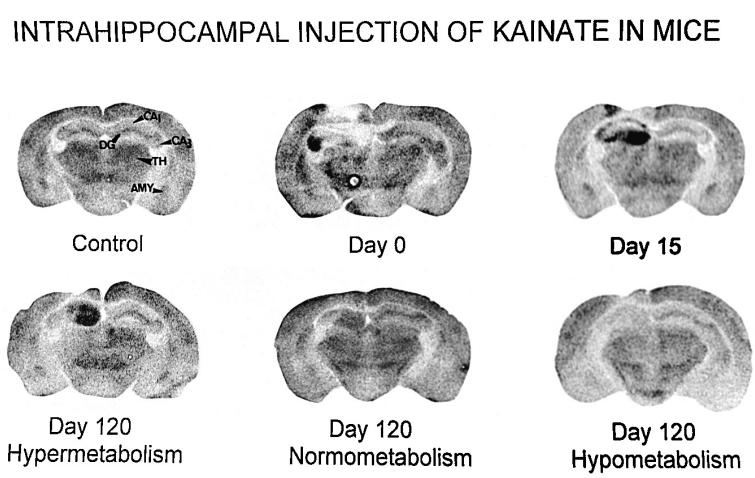


Fig. 1. [^{14}C]2DG autoradiographic sections taken at the level of the dorsal hippocampus of a control mouse and mice receiving a unilateral injection of kainate in the right hippocampus (left side on the figure) and studied at day 0 (12–15 h after the injection), day 15 and day 120 after kainate. Compared to the control animal and to the contralateral side, there was a decrease in grain density on the injection side in CA1 and an increase in grain density on the side of injection in CA3 at day 0, while the increase in grain density occurred in both CA1 and CA3 pyramidal cell layers of the injected side at day 15. At day 120, most pyramidal cells have disappeared in the pyramidal cell layers of the injected hippocampus while the granule cells of the dentate gyrus (DG) invade all the space corresponding to CA1. The dentate gyrus shows either an increase (also seen at day 15), a decrease (also seen at day 0) or no change in grain density compared to the control animal or to the contralateral side. In the other brain regions, the changes in metabolic activity were not large enough to be apparent on the autoradiograms, except for a zone of hyper- and a zone of hypoactivity apparent at day 0 and day 15 in the cortex located directly over the site of kainate injection. Abbreviations: TH: thalamus, AMY: amygdala.

two-phase process is underlying the direct excitotoxic effect of kainate during the first 2 weeks and is followed by long-term lesions that are likely to reflect the repetition of the seizures [5,6].

4.1. Evolution of cerebral functional activity during the process of hippocampal sclerosis

At day 0, at 12–15 h after the onset of SE, the location of metabolic changes induced by intrahippocampal kainate is similar to the distribution of metabolic decreases in glucose utilization occurring in the late phase of seizures induced by the acute i.p. [3] or intrastriatal injection of kainate [47]. Indeed, at 1–4 days after the i.p. administration of kainate, metabolic levels are reduced, mainly in all neocortical areas while glucose utilization remains high in the structures in which spikes are still recorded as is the case in the hippocampus and the amygdala [3]. Likewise, at 7 days after the intrastriatal injection of kainate, glucose utilization is reduced in the cerebral cortex, hippocampus, thalamus, striatum and substantia nigra pars reticulata [47].

At this early stage preceding the process of hippocampal sclerosis, there is a generalized bilateral metabolic decrease in all areas of the cerebral cortex and also in mamillary body. The generalized metabolic decrease of the cerebral cortex, always bilateral remains present at 15 days after the injection of kainate. As described previously, the first month after kainate-induced SE is characterized by most of the histopathological changes that will lead to hippocampal sclerosis [5,6]. The decrease in glucose utilization occurs in all parts of the limbic cerebral cortex that are directly or indirectly connected to the hippocampus. During the same period, there is only a limited change in functional activity in the other parts of the limbic system, which can be seen in the lateral septum, subdivisions of the amygdala and mamillary body and a metabolic decrease in the anterior and ventromedian thalamic nuclei. The extension of the hypometabolism beyond the hippocampus to cortical [2,17,18,20,27,33,34,40], thalamic [2,19,20,22] and amygdalar territories [27,46] has often been shown in patients with MTLE, i.e., when the disease has already fully developed. Bilateral cortical hypometabolism similar to the one reported in the present study has also been occasionally reported in human MTLE [2,22]. Thus, the extension of the hypometabolic zone beyond the hippocampus appears to occur in the present mouse model already during the early stages characterizing the ongoing process of hippocampal sclerosis. The exact origin of the extrahippocampal hypometabolism cannot be clearly established. Since there is no apparent neurodegeneration outside of the hippocampus, a deafferentation mechanism is the most likely hypothesis underlying the extrahippocampal hypometabolism.

The metabolic changes occurring in the various hippocampal subfields undergo a characteristic pattern of evolution. Indeed, at day 0, there is a metabolic decrease in the CA1 area, site of the injection while the CA3 region, very rich also in kainate receptors [7,45] is largely hypermetabolic. At day 15, at the onset of the phase of rapid neurodegeneration, both CA1 and CA3 become hypermetabolic which most likely reflects the active process of seizure-related excitotoxic cell death occurring as a result of the ongoing seizure activity in those cells at this time. Later on, at day 30, both layers are almost totally deprived of cell bodies and glucose utilization is no longer measurable in pyramidal cell layers. Conversely, the granule cell layer of the dentate gyrus undergoes fluctuating levels of functional activity. At all times after kainate injection, we could record increases, decreases or no change in metabolic activity. These differences most probably reflect the variable percentage of seizure activity during the 2DG experimental procedure. Indeed, these mice exhibit irregular episodes of continuous or subcontinuous seizure activity, and interictal spiking that can be recorded by means of cortical EEG monitoring. However, as seen in animals where cortical and intrahippocampal electrical activity were recorded simultaneously, most of the seizures originate in the hippocampus and have no cortical expression which renders the evaluation of their occurrence very difficult unless the mice are equipped with intrahippocampal recording electrodes (Riban, unpublished data). This was not done in the present study because we wanted to avoid any possible interference with measurements of metabolic activity, especially at the level of the hippocampus. Therefore, as described in human MTLE, the increases in functional activity in the dentate gyrus could reflect a predominance of ictal activity, decreases in glucose utilization a predominance of postictal or interictal activity while normometabolism could reflect an alternation of ictal and interictal episodes during the course of the 2DG experiment [15,32]. However, the fluctuations we record in the dentate gyrus could also partly reflect interictal spiking activity which has been shown to increase metabolic activity as well the production of immediate early genes [4,18,33].

Finally, this period of hippocampal neurodegeneration is also characterized by the decrease in glucose utilization in the substantia nigra pars compacta and in the sensorimotor cortex at days 0 and 15. The metabolic decreases in these two parts of the motor system may reflect the lack of motor components in the seizures occurring in mice subjected to an intrahippocampal injection of kainate.

4.2. Cerebral functional activity during the late phase of the epilepsy

At 30 and 120 days after kainate injection to mice, we record only discrete metabolic changes. These are an ipsilateral metabolic decrease in the entorhinal cortex, the anterior and ventromedian thalamic nuclei plus the ipsilateral hypo-, hyper- or normometabolic levels in the dentate gyrus. These data correlate quite well with the human

observations that report ipsilateral hypometabolism extending beyond the seizure focus to the ipsilateral temporal cortex [2,17,19,20,27,33,34,37,40,41,44], thalamus [2,19,20,22] and amygdala [27,46]. In humans, there is a marked asymmetry of the hypometabolism in the entorhinal, temporo-polar and anterolateral temporal cortex. In the present model, the more limited extension of hypometabolism probably reflects the restricted diffusion of the seizures outside of the hippocampus. Furthermore, the lack of change in glucose utilization in the components of the circuitry of the remote control of seizures which connects the substantia nigra pars reticulata to the caudate nucleus is also in favor of the rather localized seizure focus induced by the intrahippocampal injection of KA in mice. Indeed, in the present model there is no involvement of either the direct pathway or the indirect pathway which relays in the subthalamic nucleus and globus pallidus [1,10,11,16] while these two pathways are actively involved in models of generalized seizure [10,11,16].

In mice treated with KA and studied at 120 days after SE, we recorded bilateral metabolic increases in the central amygdala, anterior hypothalamus and mamillary body. In humans, although hypometabolism has been sometimes shown to spread contralaterally to the seizure focus [2,22], to our knowledge, there is no report of hyper- and hypometabolism occurring at the same time in different brain areas in MTLE. The hypermetabolism recorded bilaterally in the central amygdala and mamillary body may be partly independent of the seizure circuitry and rather reflect the aggressivity and anxiety generated in the animals by repetitive seizure activity. Indeed the chronically epileptic animals are not easy to handle and exhibit clear signs of anxiety and aggressivity. The amygdala is the site of regulation of anxiety and aggressivity [8] and the mamillary body belong to the Papez circuit also known to regulate emotions and anxiety. These changes also relate to the bilateral functional increases in the hypothalamus since the hippocampal formation and its principal projection system, the fornix, provide one of the main pathways by which impulses from the cortex reach the hypothalamus [29].

Thus, finally it appears that in the present model of MTLE, the ipsilateral hypometabolism most likely related to seizure activity is rather limited, fluctuating in the dentate gyrus and extending beyond the damaged hippocampus to the entorhinal cortex and thalamus, as previously shown in humans [2,17,19,20,22,27,33,34,40,46]. Thus, it appears that the metabolic decreases recorded in the present study reflect, as in humans, both a lesional and a functional origin [31]. Conversely, the bilateral metabolic increases most probably relate to the regulation of other functions not involving the seizure circuit per se.

In conclusion, the present study represents a further validation of our model of human MTLE in mice receiving a unilateral hippocampal injection of kainate. The initial process of hippocampal neurodegeneration largely involves bilateral cortical reactivity and a limited number of limbic forebrain and motor structures. When hippocampal sclerosis has fully developed, the extent of hypometabolism is limited to regions directly connected to the damaged hippocampus and most likely involved in the new hyperexcitable circuit of limbic seizures.

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References

[1] G. Alexander, M. Crutcher, Functional architecture of basal ganglia circuits neural substrates of parallel processing, Trends Neurosci. 13 (1990) 266–271.
[2] S. Arnold, G. Schlaug, H. Niemann, A. Ebner, H. Lüders, O. Witte, R.J. Seitz, Topography of interictal glucose hypometabolism in unilateral mesiotemporal epilepsy, Neurology 46 (1996) 1422–1430.
[3] Y. Ben-Ari, E. Tremblay, D. Riche, G. Ghilini, R. Naquet, Electrographic, clinical and pathological alterations following systemic administration of kainic acid, bicuculline or pentylenetetrazole: metabolic mapping using the deoxyglucose method with special reference to the pathology of epilepsy, Neuroscience 6 (1981) 1361–1391.
[4] R.G. Bittar, F. Andermann, A. Olivier, F. Dubeau, S.O. Dumoulin, G.B. Pike, D.C. Reutens, Interictal spikes increase cerebral glucose metabolism and blood flow: a PET study, Epilepsia 40 (1999) 170–178.
[5] V. Bouilleret, V. Ridoux, A. Depaulis, C. Marescaux, A. Nehlig, G. Le Gal la Salle, Recurrent seizures and hippocampal sclerosis following intrahippocampal kainate injection in adult mice: EEG, histopathology and synaptic reorganization similar to mesial temporal lobe epilepsy, Neuroscience 89 (1999) 717–729.
[6] V. Bouilleret, F. Loup, T. Kiener, C. Marescaux, J.M. Fritschy, Early loss of interneurons and delayed subunit-specific changes of $GABA_A$-receptor expression in a mouse model of mesial temporal lobe epilepsy (MTLE), J. Comp. Neurol., 1999, submitted.
[7] C.W. Cotman, D.T. Monaghan, Anatomical organization of excitatory amino acid receptors and their properties, in: R. Schwarz, Y. Ben Ari (Eds.), Excitatory Amino Acids and Epilepsy, Plenum, New York, 1985, pp. 237–252.
[8] M. Davis, The role of amygdala in fear and anxiety, Annu. Rev. Neurosci. 15 (1992) 353–375.
[9] N.C. De Lanerolle, J.H. Kim, R.J. Robbins, D.D. Spencer, Hippocampal interneuron loss and plasticity in human temporal lobe epilepsy, Brain Res. 495 (1989) 387–395.
[10] A. Depaulis, M. Vergnes, C. Marescaux, Endogenous control of epilepsy: the nigral inhibitory system, Prog. Neurobiol. 42 (1994) 33–55.
[11] C. Deransart, L. Vercueil, C. Marescaux, A. Depaulis, The role of basal ganglia in the control of generalized absence seizures, Epilepsy Res. 32 (1998) 213–223.
[12] J. Engel Jr., Update on surgical treatment of the epilepsies: summary of the Second International Palm Desert Conference on the Surgical Treatment of the Epilepsies (1992), Neurology 43 (1993) 1612–1617.
[13] J. Engel Jr., Critical evaluation of animal models for localisation-related epilepsies, Ital. J. Neurol. Sci. 16 (1995) 9–16.
[14] M.A. Falconer, E. Serafetinides, J.A.N. Corsellis, Etiology and pathogenesis of temporal lobe epilepsy, Arch. Neurol. 10 (1964) 233–248.

[15] G. Franck, B. Sadzot, E. Salmon, J.C. Depresseux, T. Grisar, J.M. Peters, M. Guillaume, L. Quaglia, G. Delfiore, D. Lamotte, Regional cerebral blood flow and metabolic rates in human focal epilepsy and status epilepticus, in: A.V. Delgado-Escueta, A.A. Ward, D.M. Woodbury, R.J. Porter (Eds.), Advances in Neurology, Vol. 44, Raven Press, New York, 1986, pp. 935–948.
[16] D.S. Garant, K. Gale, Substantia nigra-mediated anticonvulsant actions: role of nigral output pathways, Exp. Neurol. 97 (1987) 143–159.
[17] M. Hajek, A. Antonini, K.L. Leenders, H.G. Wieser, Mesiobasal versus lateral temporal lobe epilepsy: metabolic differences in the temporal lobe shown by interictal ^{18}F-FDG positron emission tomography, Neurology 43 (1993) 79–86.
[18] A. Handforth, D.M. Finch, R. Peters, A.M. Tan, D.M. Treiman, Interictal spiking increases 2-deoxy[^{14}C]glucose uptake and c-*fos*-like reactivity, Ann. Neurol. 35 (1994) 724–731.
[19] T.R. Henry, J.C. Mazziotta, J. Engel Jr., P.D. Christenson, J.X. Zhang, M.E. Phelps, D.E. Kuhl, Quantifying interictal metabolic activity in human temporal lobe epilepsy, J. Cereb. Blood Flow Metab. 10 (1990) 748–757.
[20] T.R. Henry, J.C. Mazziotta, J. Engel Jr., Interictal metabolic anatomy of mesial temporal lobe epilepsy, Arch. Neurol. 50 (1993) 582–589.
[21] C.R. Houser, Granule cell dispersion in the dentate gyrus of humans with temporal lobe epilepsy, Brain Res. 535 (1990) 192–204.
[22] N. Khan, K.L. Leenders, M. Hajek, P. Maguire, J. Missimer, H.G. Wieser, Thalamic glucose metabolism in temporal lobe epilepsy measured with ^{18}F-FDG positron emission tomography (PET), Epilepsy Res. 28 (1997) 233–243.
[23] D. Lurton, L. Sundstrom, C. Brana, B. Bloch, A. Rougier, Possible mechanisms inducing granule cell dispersion in humans with temporal lobe epilepsy, Epilepsy Res. 26 (1997) 351–361.
[24] J.H. Margerison, J.A.N. Corsellis, Epilepsy and the temporal lobe, Brain 89 (1966) 499–530.
[25] G.W. Mathern, Hippocampal sclerosis, in: J. Engel, Jr., T.A. Pedley (Eds.), Epilepsy. A Comprehensive Textbook, Lippincott Raven Publ., Philadelphia, 1997, pp. 133–155.
[26] G.W. Mathern, T.L. Babb, J.K. Pretorius, J.P. Leite, Reactive synaptogenesis and neuron densities for neuropeptide Y, somatostatin, and glutamate decarboxylase immunoreactivity in the epileptogenic human fascia dentata, J. Neurosci. 15 (1995) 3990–4004.
[27] M. Mikkonen, H. Soininen, R. Kälviäinen, T. Tapiola, A. Ylinen, M. Vapalahti, L. Paljärvi, A. Pitkänen, Remodeling of neuronal circuitries in human temporal lobe epilepsy: increased expression of highly polysialylated neural cell adhesion molecule in the hippocampus and the entorhinal cortex, Ann. Neurol. 44 (1998) 923–934.
[28] A. Nehlig, A. Pereira de Vasconcelos, S. Boyet, Quantitative autoradiographic measurement of local cerebral glucose utilization in freely moving rats during postnatal development, J. Neurosci. 8 (1988) 2321–2333.
[29] J.W. Papez, A proposed mechanism of emotion, Arch. Neurol. Psychiatry 38 (1937) 725–743.
[30] R.A. Radtke, M.W. Hanson, J.M. Hoffmann, B.J. Crain, T.S. Walczak, D.V. Lewis, C. Beam, R.E. Coleman, A.H. Friedman, Temporal lobe hypometabolism on PET: predictor of seizure control after temporal lobectomy, Neurology 43 (1993) 1088–1092.
[31] A. Rougier, D. Lurton, B. El Bahh, V. Lespinet, A.M. Bidabé, M. Guillot, J.M. Caillé, Bilateral decrease in interictal hippocampal blood flow in unilateral mesiotemporal epilepsy, J. Neurosurg. 90 (1999) 282–288.
[32] C.C. Rowe, S.F. Berkovic, M.C. Austin, W.J. McKay, P.F. Bladin, Patterns of postictal cerebral blood flow in temporal lobe epilepsy: qualitative and quantitative analysis, Neurology 41 (1991) 1096–1103.
[33] E. Rubin, V. Dhawan, J.R. Moeller, S. Takikawa, D.R. Labar, N. Schaul, W.B. Barr, D. Eidelberg, Cerebral metabolic topography in unilateral temporal lobe epilepsy, Neurology 45 (1995) 2212–2223.
[34] J.C. Sackellares, G.J. Siegel, B.W. Abou-Khalil, T.W. Hood, S. Gilman, P.E. McKeever, R.D. Hichwa, G.D. Hutchins, Differences between lateral and mesial temporal metabolism interictally in epilepsy of mesial temporal origin, Neurology 40 (1990) 1420–1426.
[35] B. Sadzot, R.M. Debets, G. Delfiore, C.W. van Huffelen, A.C. van Veelen, C. Degueldre, D. Comar, G. Franck, Decrease of ^{11}C-flumazenil binding is more localized than glucose hypometabolism in patients with TLE studied by PET, Neurology 44 (Suppl. 2) (1994) A351–A352.
[36] I. Savic, M. Ingvar, S. Stone-Elander, Comparison of (^{11}C)flumazenil and (^{18}F)FDG as PET markers of epileptic foci, J. Neurol. Neurosurg. Psychiatry 56 (1993) 615–621.
[37] F. Semah, M. Baulac, D. Hasboun, V. Frouin, J.F. Mangin, S. Papageorgiou, A. Leroy-Willig, J. Philippon, D. Laplane, Y. Samson, Is interictal temporal hypometabolism related to mesial temporal sclerosis? A positron emission tomography/magnetic resonance imaging confrontation, Epilepsia 36 (1995) 447–456.
[38] C. Shin, Functional neuropathology of temporal lobe epilepsy, in: G.D. Cascino, C.R. Jack (Eds.), Neuroimaging in Epilepsy: Principles and Practice, Butterworth–Heinemann, Boston, 1996, pp. 65–71.
[39] L. Sokoloff, M. Reivich, C. Kennedy, M.H. DesRosiers, C.S. Patlak, K.D. Pettigrew, O. Sakurada, M. Shinohara, The [^{14}C]deoxyglucose method for the measurement of local cerebral glucose utilization: theory, procedure, and normal values in the conscious and anesthetized albino rat, J. Neurochem. 28 (1977) 897–916.
[40] M.R. Sperling, R.C. Gur, A. Alavi, R.E. Gur, S. Resnick, M.J. O'Connor, M. Reivich, Subcortical metabolic alterations in partial epilepsy, Epilepsia 31 (1990) 145–155.
[41] H. Stefan, G. Pawlik, H.G. Bocher-Schwarz, H.J. Biersack, W. Burr, H. Penin, W.D. Heiss, Functional and morphological abnormalities in temporal lobe epilepsy: a comparison of interictal and ictal EEG, CT, MRI, SPECT and PET, J. Neurol. 234 (1987) 377–384.
[42] F. Suzuki, M.P. Junier, D. Guilhem, J.C. Sorensen, B. Onteniente, Morphogenetic effect of kainate on adult hippocampal neurons associated with a prolonged expression of brain-derived neurotrophic factor, Neuroscience 64 (1995) 665–674.
[43] W.H. Theodore, D. Katz, C. Kufta, S. Sato, N. Patronas, P. Smothers, E. Bromfield, Pathology of temporal lobe foci: correlation with CT, MRI and PET, Neurology 40 (1990) 797–803.
[44] W.H. Theodore, S. Sato, C. Kufta, M.B. Balish, E. Bromfield, D.B. Leiderman, Temporal lobectomy for uncontrolled seizures: the role of positron emission tomography, Ann. Neurol. 32 (1992) 789–794.
[45] E. Tremblay, A. Represa, Y. Ben-Ari, Autoradiographic localization of kainic acid binding sites in the human hippocampus, Brain Res. 343 (1985) 378–382.
[46] W. Van Paesschen, A. Connelly, C.L. Johnson, J.S. Duncan, The amygdala and intractable temporal lobe epilepsy: a quantitative magnetic resonance imaging study, Neurology 47 (1996) 1021–1031.
[47] G.F. Wooten, R.C. Collins, Regional brain glucose utilization following intrastriatal injections of kainic acid, Brain Res. 201 (1980) 173–184.